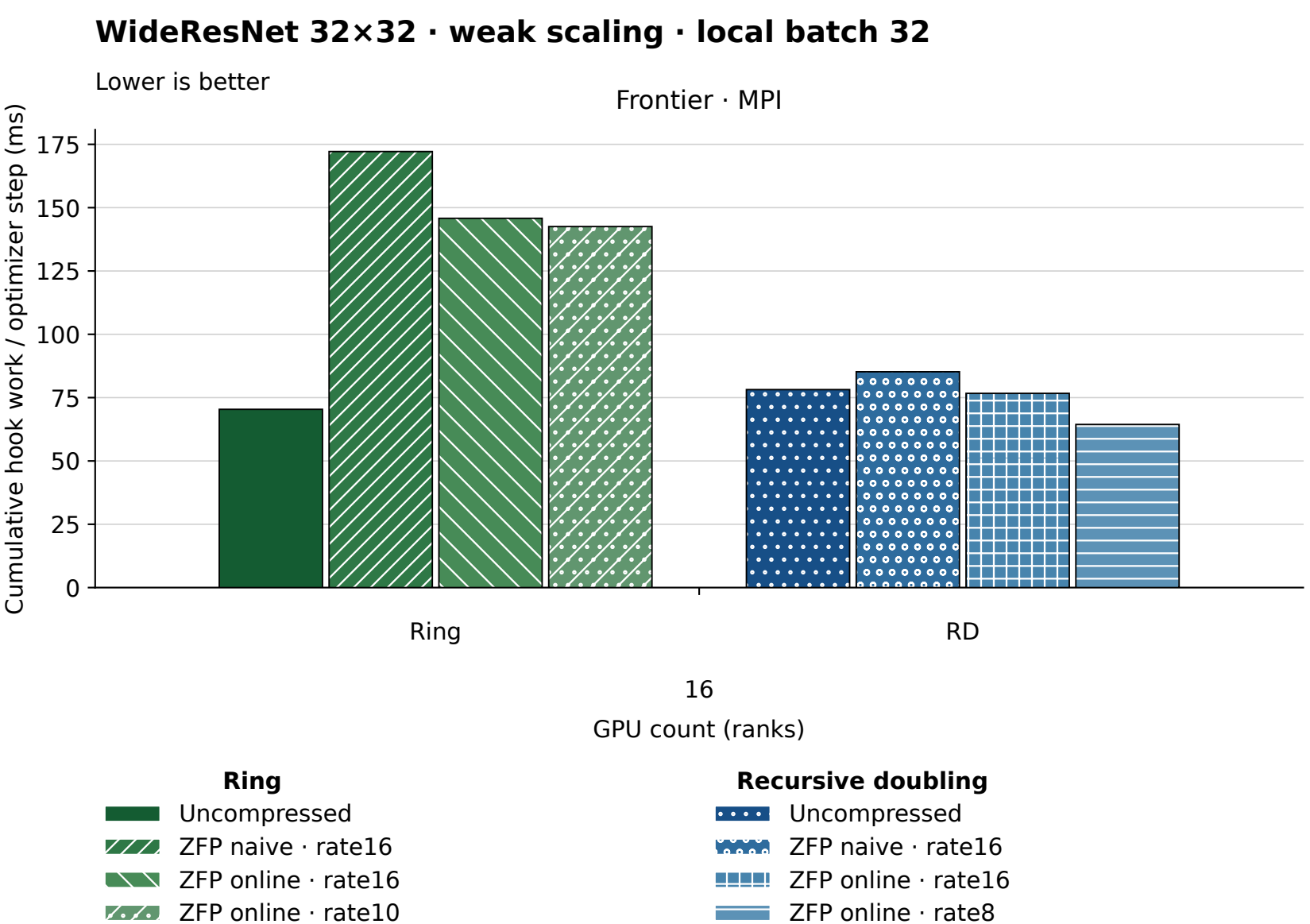

WideResNet 32×32 · weak scaling · local batch 32
Lower is better
Frontier · MPI
Cumulative hook work / optimizer step (ms)
175
150
125
100
75
50
25
0
Ring
RD
16
GPU count (ranks)
Ring
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate10
Recursive doubling
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate